FastRW: An Efficient Random Walk Method for Steady-State Thermal Analysis

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Key Results

FastRW accelerates Feynman–Kac-based random walk thermal simulation through exact truncation error analysis and prior/posterior estimation, achieving **over 6.5× speedup** over prior state-of-the-art methods without sacrificing accuracy.

Motivation

Accurate thermal simulation is critical for modern IC design with increasing power density. Random walk methods based on the Feynman–Kac formula enable efficient **localized** temperature estimation without solving the global field. However:

- In practical scenarios **without Dirichlet boundaries**, long random walk paths are needed for convergence.
- Existing methods select truncation thresholds **empirically** with no principled error guarantee.
- FastRW (ours): provides quantitative truncation error analysis and uses prior/posterior estimation to dramatically shorten paths.

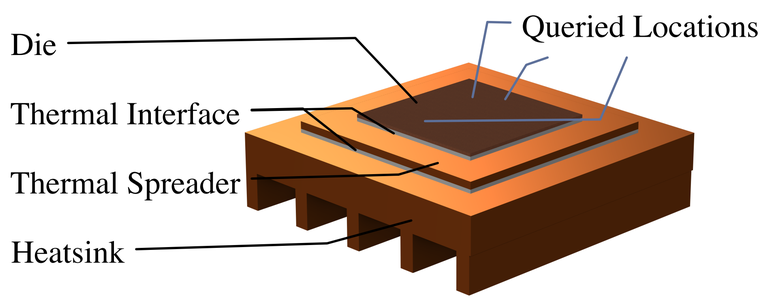


Figure 1. In many thermal design scenarios, only temperatures at a few queried hotspot locations are needed — ideal for random walk methods.

Feynman–Kac Representation

The temperature at $\boldsymbol{x}$ is expressed via the Feynman–Kac formula as an expectation over reflecting Brownian motion (SRBM) paths:

$$T(\boldsymbol{x}) = \mathbb{E}\left\{\int_0^\infty \hat{e}_c(t)\, f(X_t)\,\mathrm{d}t\right\} + \mathbb{E}\left\{\hat{e}_c(t_D)\,\phi_D(X_{t_D})\right\} + \mathbb{E}\left\{\int_0^\infty \hat{e}_c(t)\,\phi_N(X_t)\,\mathrm{d}L(t)\right\} + \mathbb{E}\left\{\int_0^\infty \hat{e}_c(t)\,\phi_R(X_t)\,\mathrm{d}L(t)\right\}, \tag{1}$$

where $\hat{e}_c(t) = \exp\left(-\int_0^t c(X_s)\,\mathrm{d}L(s)\right)$ and $L(t)$ is the local time at the boundary.

Truncation Error Analysis

For a threshold $\Lambda \in (0,1)$, we truncate the path at t_Λ when $\hat{e}_c(t) \le \Lambda$. The key result:

$$\mathbb{E}\left[\mathbb{P}_{t_\Lambda}^\infty\right] = \hat{e}_c(t_\Lambda)\,T\left(X_{t_\Lambda}\right) = \Lambda\,T\left(X_{t_\Lambda}\right). \tag{2}$$

Corollary: The expected truncation error equals the **weighted temperature** at the truncation location X_t. This provides a principled guideline for choosing Λ.

Single-Point Acceleration via Prior Estimation

We use a coarse FEM prior $\tilde{T}(\boldsymbol{x}) = T(\boldsymbol{x}) + \epsilon$ to compensate the truncation error:

$$T_i(\boldsymbol{x}) = \mathbb{P}_0^t + \hat{e}_c(t)\,\tilde{T}(\boldsymbol{x}) = \mathbb{P}_0^t + \mathbb{E}[\mathbb{P}_t^\infty] + \hat{e}_c(t)\,\epsilon. \tag{3}$$

Since $\epsilon \ll T(\boldsymbol{x})$, the threshold Λ can be **350× larger**, cutting path length by more than half.

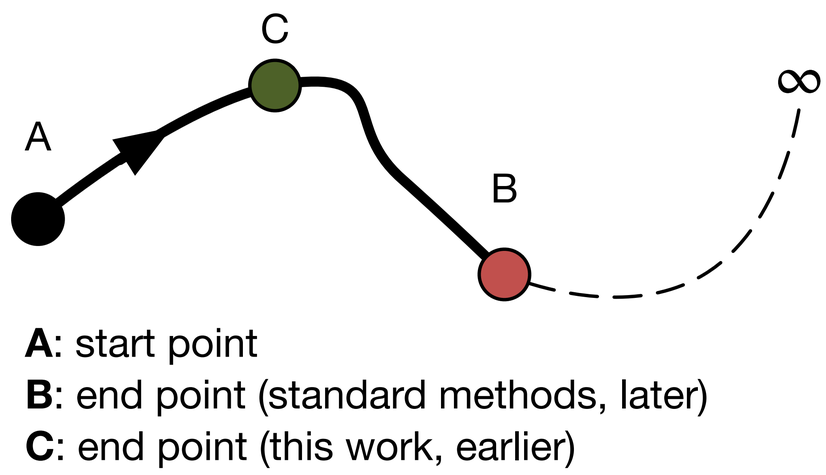


Figure 2. Standard methods compute the full path A→B. FastRW truncates at C and estimates the remaining contribution using the prior.

Multi-Point Acceleration via Posterior Observation

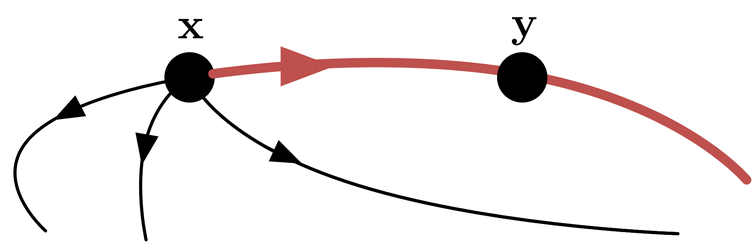
When estimating temperatures at M points $\{\boldsymbol{x}_q\}_{q=1}^M$, paths starting at $\boldsymbol{x}$ that visit $\boldsymbol{y}$ yield a noisy linear observation:

$$b_k = \mathbf{T}_{i_k} - \alpha_k\,\mathbf{T}_{j_k} + \eta_k, \quad \text{stacked as } \mathbf{B} = \boldsymbol{A}\,\mathbf{T} + \boldsymbol{\eta}. \tag{4}$$

With Bayesian inference (Gaussian prior from single-point estimates):

$$\hat{\mathbf{T}} = \boldsymbol{\Sigma}_{\text{post}}\left(\boldsymbol{A}^\top \boldsymbol{R}^{-1}\mathbf{B} + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0\right), \tag{5}$$

cross-point information further reduces the number of paths needed.



Experimental Setup

- **Test cases:** 3D chip model (3 layers: source-free / heat-source / source-free), 3 thickness configs, 3 power maps (`power6`, `4-core`, `16-core`).
- **Baseline:** PIRW [1], a Feynman–Kac method with standard truncation ($\Lambda = 0.0001$).
- **Hardware:** 2× Intel Xeon Silver 4214 (24 cores), 128 GB RAM.

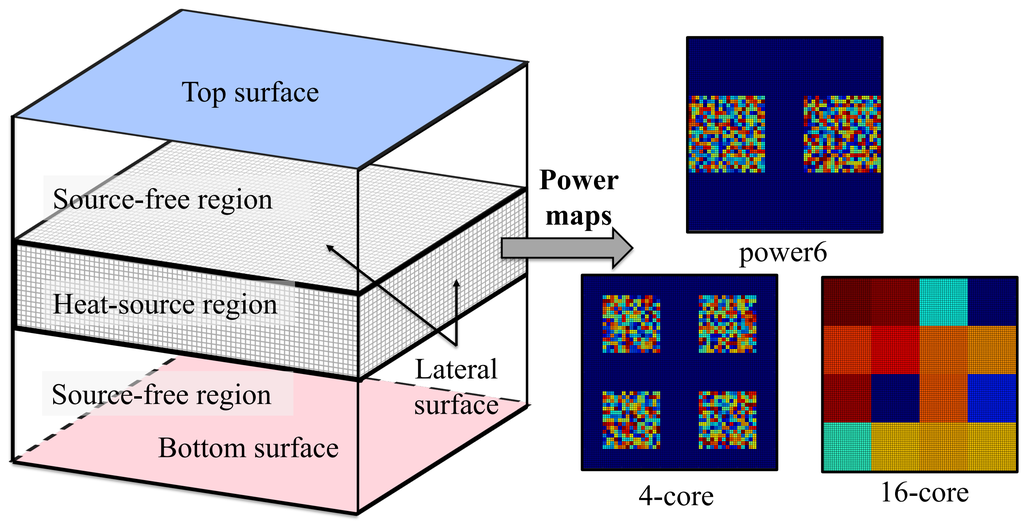


Figure 4. Test case geometry and power map configurations.

Single-Point Results

Table 1. Temperature estimation comparison. FastRW uses $\Lambda = 0.03$ with coarse FEM prior.

Case	Power Map	PIRW [1] Error/K	PIRW [1] StdDev/K	PIRW [1] Steps	PIRW [1] Paths	FastRW Error/K	FastRW StdDev/K	FastRW Steps	FastRW Paths	Speedup
1	`power6`	0.38	0.47	5.79e6	1000	**0.36**	**0.45**	2.20e6	400	**6.58×**
2	`4-core`	0.43	0.55	5.83e6	1000	**0.42**	**0.54**	2.22e6	400	**6.57×**
3	`16-core`	0.38	0.46	1.03e7	1000	**0.32**	**0.41**	3.93e6	400	**6.55×**

- FastRW achieves **6.5×+ speedup** with **improved accuracy** across all cases.
- Path length reduced by >50%; number of required paths reduced from 1000 to 400.

Multi-Point Acceleration

Table 2. Multi-point estimation (Case 1, `power6`).

Group Size	Error(K)	StdDev(K)	$\hat{N}$	Speedup
1	0.356	0.446	1000	1.00×
4	0.292	0.379	1385	1.39×
16	0.221	0.281	2519	**2.52×**

Table 3. Efficiency analysis: time per point (Case 1).

Group Size	Prior(s)	RW(s)	Posterior(s)	Extra Cost(%)
1	0.28	15.41	0.00	1.82
4	0.28	15.41	0.02	1.95
16	0.28	15.41	0.61	5.78

- Larger group sizes yield more cross-point observations ⇒ additional **2.52×** acceleration.
- Prior + posterior overhead is only **5.78%** even at group size 16.

Effect of Prior Accuracy

Table 4. Better FEM prior enables larger Λ and shorter paths.

Prior DoF	Prior Max-Error (K)	Random Walk Threshold Λ	Random Walk Steps	Random Walk Time (s)
1875	1.93	0.01	2.89e6	20.2
4800	0.88	0.02	2.45e6	17.1
12500	0.59	0.03	2.20e6	15.4

More accurate priors (finer FEM) reduce truncation error, allowing larger Λ and further runtime savings.

Conclusion

FastRW accelerates thermal analysis by:

1. **Exact truncation error analysis** — the expected error equals the weighted temperature at the truncation point.
2. **Prior estimation** — coarse FEM priors shorten paths by >50%, yielding **6.5×** single-point speedup.
3. **Posterior correction** — cross-point Bayesian inference provides up to **2.5×** additional acceleration with <6% overhead.

Future extensions: more accurate priors (learning-based), transient conduction, and GPU acceleration.

References

[1] Hao-Liang Yang, Yu-Cheng Zhong, Shang-Wei Peng, and Yih-Lang Chang. Practical heat conduction estimation with random walk. In *Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design*, 2022.